

Ejercicio 2:

Hacer en lápiz y papel y subir un PDF con la resolución.

Dado $h[n] = \delta[n] - \delta[n - 4]$, encontrar $y[n] = x[n] * h[n]$ para cada una de las siguientes $x[n]$:

a) $x[n] = \cos(\omega_0 \cdot n \cdot T_s)$. Expresar la respuesta como un único coseno de la forma $A \cos(\omega_0 \cdot n \cdot T_s + \varphi)$.

b) $x[n] = (1/2)^n u[n]$.

c) $x[n] = u[n + 1] - u[n - 2]$.

$$h[n] = \delta[n] - \delta[n - 4]$$

ITA
de
impulso

$$y[n] = x[n] * h[n]$$

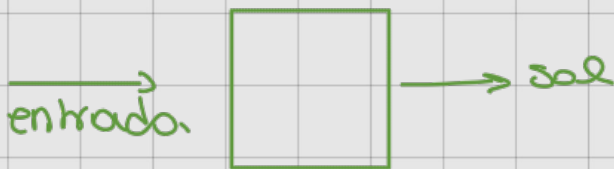
Salida

convolucionar
con una delta
desplazada,
desplazo la
entrada

usando la
Propiedad

$$y[n] = x[n] * \delta[n] - x[n] * \delta[n - 4]$$

$$y[n] = x[n] - x[n - 4]$$



Entrada:

a) $x[n] = \cos(\omega_0 n T_s)$

$$y[n] = \cos(\underbrace{\omega_0 n T_s}_{\alpha}) - \cos(\underbrace{\omega_0 (n-4) T_s}_{\beta})$$

$$\cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right)$$

$$\frac{\alpha - \beta}{2} = \frac{\omega_0 n T_s - \omega_0 (n-4) T_s}{2}$$

$$\frac{\alpha + \beta}{2} = \frac{\omega_0 n T_s + \omega_0 (n-4) T_s}{2}$$

$$\frac{\alpha - \beta}{2} = 2\omega_0 T_s$$

$$\frac{\alpha + \beta}{2} = \omega_0 n T_s - 2\omega_0 T_s$$

$$y[n] = -2 \sin(\omega_0 n T_s - 2\omega_0 T_s) \underbrace{\sin(2\omega_0 T_s)}_{\text{no dep. de } n}$$

uso $-\sin(\theta) = \cos(\theta + \frac{\pi}{2})$

$$y[n] = 2\sin(2\omega_0 T_s) \cos(\omega_0 n T_s - 2\omega_0 T_s + \frac{\pi}{2})$$

$$y[n] = A \cos(\omega_0 n T_s + \phi)$$

$$A = 2\sin(2\omega_0 T_s)$$

$$\phi = -2\omega_0 T_s + \frac{\pi}{2}$$

} modifica amplitud
y fase, no cambia
la frecuencia

b)

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$y[n] = x[n] - x[n-4]$$

$$x[n-4] = \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

$$y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

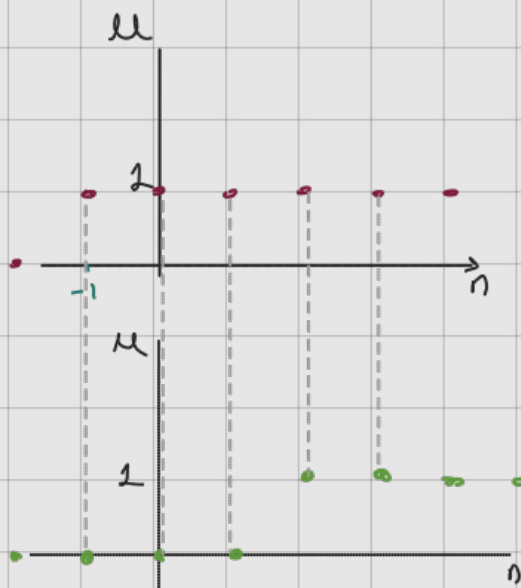
$$y[n] = \begin{cases} 0 & n < 0 \\ \left(\frac{1}{2}\right)^n & 0 \leq n < 4 \\ \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4} & n \geq 4 \end{cases}$$

c) $x[n] = u[n+1] - u[n-2]$

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

$\begin{cases} u[n+1] \rightarrow \text{activa en } n=-1 \\ u[n-2] \rightarrow \text{activa en } n=2 \end{cases}$
 Si los resto quedan
 activos solo $n=-1, 0, 1$

$$x[-1] = x[0] = x[1] = 1$$



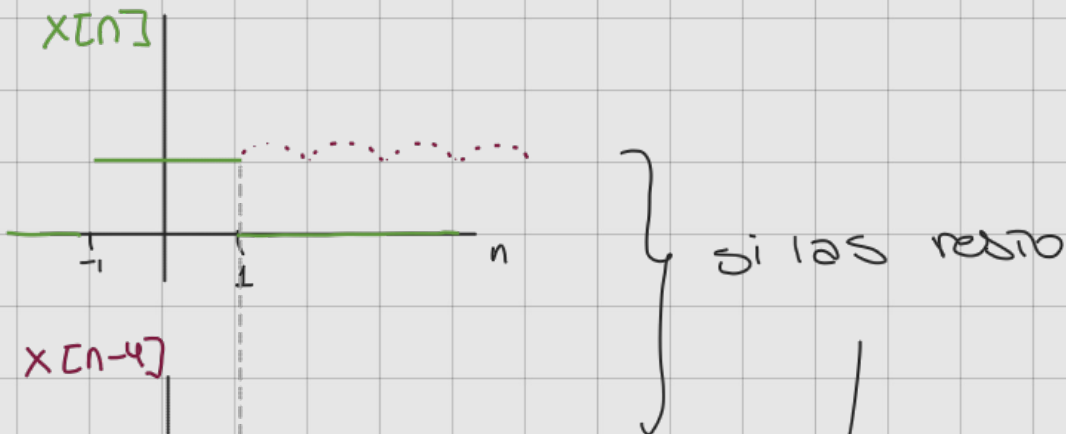
$$y[n] = x[n] - x[n-4]$$

$$x[n] = u[n+1] - u[n-2]$$

$$x[n-4] = u[n-3] - u[n-6]$$

$$y[n] = u[n+1] - u[n-2] - [u[n-3] - u[n-6]]$$

$$y[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6]$$



$$y[n] = x[n] - x[n-4]$$

